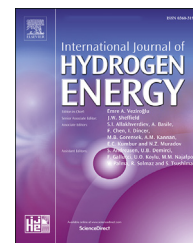


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Hydrogen Economy Assessment & Resource Tool (HEART): A python-based tool for ASEAN H₂ roadmap study

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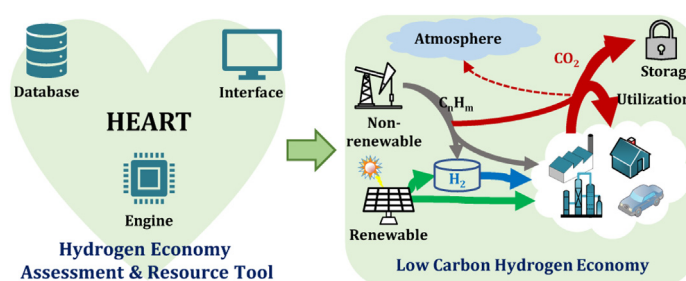
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HIGHLIGHTS

- A Python-based tool is developed for hydrogen economy planning and optimization.
- The framework, workflow, and capabilities of HEART is introduced.
- Hydrogen supply chains for multiple sectors are planned for an ASEAN case study.

GRAPHICAL ABSTRACT



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ABSTRACT

In the past decades, H₂ has attracted significant attention as a potentially low, zero, or negative-emissions fuel depending on how it is produced. However, how H₂ will evolve in terms of its production, demand, and transport is not very clear. To help fill this gap, we developed a Python-based tool called the Hydrogen Economy Assessment & Resource Tool (HEART), specifically focused on the Association of Southeast Asian Nations (ASEAN) region. The tool consists of three parts: a user-friendly GUI, a default database consisting of costs and emissions for various supply/demand pathways, and an analysis and optimization engine. Users can use the default data and/or user inputs to run case studies for insightful results. The analysis and optimization engine can carry out techno-environmental analysis for H₂ production and H₂ transport, project H₂ demand, and carry out the optimization and planning of H₂ supply chain over a long-term period. In this paper, we introduce the framework of HEART, the workflow of various projects, and show the

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Optimization
MILP

capabilities and strengths of HEART through a case study. The case study indicates the tool can provides many useful results, like the landed cost and carbon footprint of H₂. Most importantly, by using HEART, H₂ supply chain pathways to meet projected demand can be obtained and optimized for total cost or total emissions.

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Introduction

In view of the consequences of rising atmospheric CO₂ concentration, the world has been seeking low-carbon technologies in recent years. A recent study analyzed the climate change mitigation ambition of 327 European cities [1]. The analysis revealed that 78% of cities have a mitigation plant with targets with an average of 47% reduction in greenhouse gas (GHG) emissions. Hydrogen, as a low-emission fuel, is receiving significant attention worldwide [2,3]. Many studies have been conducted to find roadmap alternatives for H₂ economy that help answer questions like how to produce H₂ [4], how to transport H₂ [5].

Hydrogen can be produced using a range of energy sources and technologies [6]. The cost of and emissions from H₂ production by different means can be significantly different [7]. Two recent reviews about H₂ production have been contributed by Nazir et al. [8] and Dawood et al. [9]. Two main technologies for producing hydrogen from fossil fuels are hydrocarbon reforming and pyrolysis [6]. Hydrocarbon reforming is a process where a hydrocarbon feedstock (e.g. natural gas, naphtha, ...) is converted into hydrogen. The leading technologies include steam reforming [10] (generally referred as steam methane reforming, SMR), autothermal reforming (ATR), and partial oxidation (POX). Hydrocarbon pyrolysis [11] is not widely applied to produce hydrogen in the industries today. However, it has the potential to be one of the efficient “low-carbon” technologies, since solid carbon rather than CO₂ is produced as the by-product of this hydrogen production process. Kerscher et al. [12] stated the carbon footprint of hydrogen via beam plasma methane pyrolysis is between 1.9 kg/CO₂ eq./kg H₂ and 6.4 kg CO₂/kg H₂ depending on the electricity sources. Solid carbon can be used for the production of carbon nano tubes, carbon black, carbon fibers, and graphene, while a large scale of deployment of hydrocarbon pyrolysis would exceed the current market of these product. New utilization routes for solid carbon from hydrogen pyrolysis need to be developed [13]. Hydrogen production from renewable sources [14] like biomass [15,16] and water splitting using renewable electricity [17,18] are some of the promising alternatives. Hydrogen can be extracted from biomass by thermochemical and biological processes. Low-carbon hydrogen [19] can be obtained from biomass resources, if CO₂ capture and storage is deployed. The thermochemical technology mainly involves biomass pyrolysis and biomass gasification (BG), while the biological processes include bio-photolysis and fermentation. The water splitting processes consist of electrolysis, thermolysis, and photo-electrolysis. Among them, water electrolysis [20] (WE) is the

most efficient commercial process. To date, alkaline electrolysis (AEL) and proton exchange membrane electrolysis (PEM), and solid oxide electrolysis (SOEL) [21] are the three dominant technologies. Water electrolysis will play a fundamental role in the transition towards the renewable H₂ economy, as electrolyzers can convert renewable electricity into H₂ for energy storage and transport.

Hydrogen has an energy density of approximately 33.3 kWh/kg, almost three times more than diesel. However, hydrogen has extremely low volumetric energy density at atmospheric temperature and pressure: only 2.7 kWh/m³. When compressed to 700 bar, it can be increased to 1332 kWh/m³ which is just 1/10th of diesel [22]. As such, for its transport, hydrogen must be compressed, liquified [23], or embedded into other carriers (e.g. liquid organic hydrogen carrier - LOHC [24–26], ammonia [27,28], metal–organic frameworks- MOF [29]) before transportation. Different transport methods have processing requirements before delivery (e.g. compression, liquefaction, and hydrogenation.) and for converting to a useable form after reaching the demand sites (e.g. recompression, regasification, and dehydrogenation). Regional [30] and even global H₂ supply chains are being explored to help energy importing countries participate in a low-carbon H₂ economy.

Many studies have been conducted for hydrogen supply chains via various carriers. Heuser et al. [31] carried out a techno-economic analysis of a potential CO₂-free liquid hydrogen supply chain between Patagonia and Japan. Niermann et al. [24] used comprehensive simulations to examine a H₂ supply chain using various LOHC options. The results indicated that LOHC options could provide efficient and economically viable transport and storage in a H₂ economy. Anastasopoulou et al. [29] conducted techno-economic modeling to quantify cost, performance, and relationships between the system components for an MOF-based hydrogen supply chain, and compared them with those for compressed and liquified hydrogen. They also identified a few key opportunities to lower the cost by developing new MOFs with higher volumetric capacities and improved cycling stability. Wijayanta et al. [32] compared hydrogen supply chains from Norway to Japan via liquid hydrogen, methylcyclohexane (MCH), and liquid ammonia. Their results showed that ammonia with direct utilization as fuel for a combustion engine or fuel cell (without decomposition back to H₂) was best positioned for massive adoption.

Development of a H₂ economy involves a series of decisions and selections related to technology choice, when and where to construct related infrastructure, and at what capacity [5]. Parra et al. [33] concluded that technology

developers should elaborate smart strategies to reduce cost by scaling production and develop new market structures to deliver multiple low carbon applications. Maestre and Ortiz [34] systematically analyzed the challenges and prospects of full decarbonization of stationary power applications via renewable hydrogen-based strategies. Main barriers, such as lack of infrastructures, safety risks, and lack of policies, are identified. In light of the above review, it can be seen that each decision has many alternatives, which are interconnected in a manner that leads to many trade-offs. Thus, it is challenging for decision-makers to identify promising pathways from a large number of available options. Support from a versatile tool capable of analyzing all available roadmap options based on a set of techno-enviro-economic criteria can play an important role in to help the decision makers in the planning and development of a H₂ economy.

In this study, we present a python-based tool for the planning and development of a H₂ economy with focus on the ASEAN region. The full name of the tool is Hydrogen Economy Assessment & Resource Tool (HEART). The tool includes three parts: (1) a user-friendly graphical user interface (GUI), which allows users to provide inputs and interact with the tool; (2) a default database for hydrogen demand projection, production cost estimation including CO₂ capture and sequestration, available modes of H₂ transport, their cost and environmental impact; and (3) an analysis and optimization engine to analyze, optimize, and visualize H₂ cost, emissions, capital investment, etc., as a function of time. In the following, we will first introduce the framework of HEART, followed by a systematic introduction of the workflow and capabilities of the tool. Next, a demo case study is presented with conclusions.

Framework of HEART

The framework of HEART is shown in Fig. 1, which includes three parts: (1) database, (2) GUI, and (3) “engine”. The database is maintained via spreadsheet files (.csv and .xlsx extension files). The GUI is constructed by the Tkinter package in Python. The engine includes a multiple period H₂ supply chain optimization model implemented in GAMS (General Algebraic Modeling System) and techno-enviro-economic analysis tool coded in Python.

Published data are used to populate the database with default values for H₂ demand, H₂ production cost, and H₂ transport cost, as well as related GHG emissions (expressed in CO₂ equivalent). The database is made available for data processing and process simulation. Techno-enviro-economic analysis procedures [22] are systematically generalized as functions via Python. A mixed-integer linear programming (MILP) model is developed for the optimization of a H₂ supply chain, which can be either local or regional. Finally, a GUI is constructed for a user to interact with the default database, the techno-enviro-economic analysis tools, and the optimization model in order to build and run a case study.

The dataflow for a user to run a case study is shown by the blue lines in Fig. 1. One can use the tool without any new inputs if default inputs are acceptable, but the users can override any default data with inputs. After loading default data

and/or adding user data, the user can obtain sector by sector or aggregate projection of H₂ demand, H₂ production cost and emissions via various technologies, and H₂ transport cost and emissions for different carriers (compressed hydrogen - CH₂, liquid ammonia - LNH₃, liquid hydrogen - LH₂, and methylcyclohexane - MCH) using HEART. By invoking the optimization model in GAMS, one can also obtain the optimal H₂ supply chain representing minimum cost and/or emissions under different scenarios for a chosen time period. The obtained results are displayed in the GUI and stored for accessing at a later time. Before exiting the tool, one can choose to package and compress all case study inputs and outputs (results). The software also has the option to open a case study from a compressed file package to revisit, modify, and update the case study.

GUI and workflow of HEART

The homepage of the HEART graphical user interface is shown in Fig. 2, which consists of a menu bar, hyperlinks to tool tutorial and information, and entries to start a new project (case study). There are four types of projects available: H₂ Demand, H₂ Production, H₂ Transport, and Optimization. The first two are independent projects, while a H₂ Transport project automatically includes Demand and Production. An Optimization project naturally includes the other three projects, because their data will be inputs to the optimization model. More details about the GUI are shown in a demo-case-study video given in the [Supporting Information](#).

Workflow – Step 1: Basic information

The workflow of the aforementioned four project types is summarized in Fig. 3, where the dataflow from the default database to the user database is also shown. Once a new project is created (#1 in Fig. 3), HEART automatically creates all required pages, e.g., one basic information page and one H₂ production page for a H₂ Production project. One can change the project type by adding or deleting pages through the menu bar. As shown in Fig. 3, no matter what project one creates, it starts from selecting the period of analysis (start and end dates) and locations (#2). Default information is loaded from the database for all added locations, as shown by the blue line. Note that the current database only includes data for the ASEAN region. However, the HEART tool can be easily applied for any regions across the world with users' own data. Location attributes include grid electricity price, various energy source price, location factor [35], carbon tax, etc. Location factor is used to scale up/down capital cost accordingly, where the United States has a value of 1. Each attribute consists of four parameters: base value (base), reference year (year 0), and model parameters to capture the change of each attribute with time. Provisions are made for parameters to change over time via either a linear model, which assumes a fixed annual change (a per year), or a compound change model, which assumes a fixed rate of change (b % per year).

Based on these four parameters, a projection (proj) for the entire period of analysis can be obtained according to the following equation:

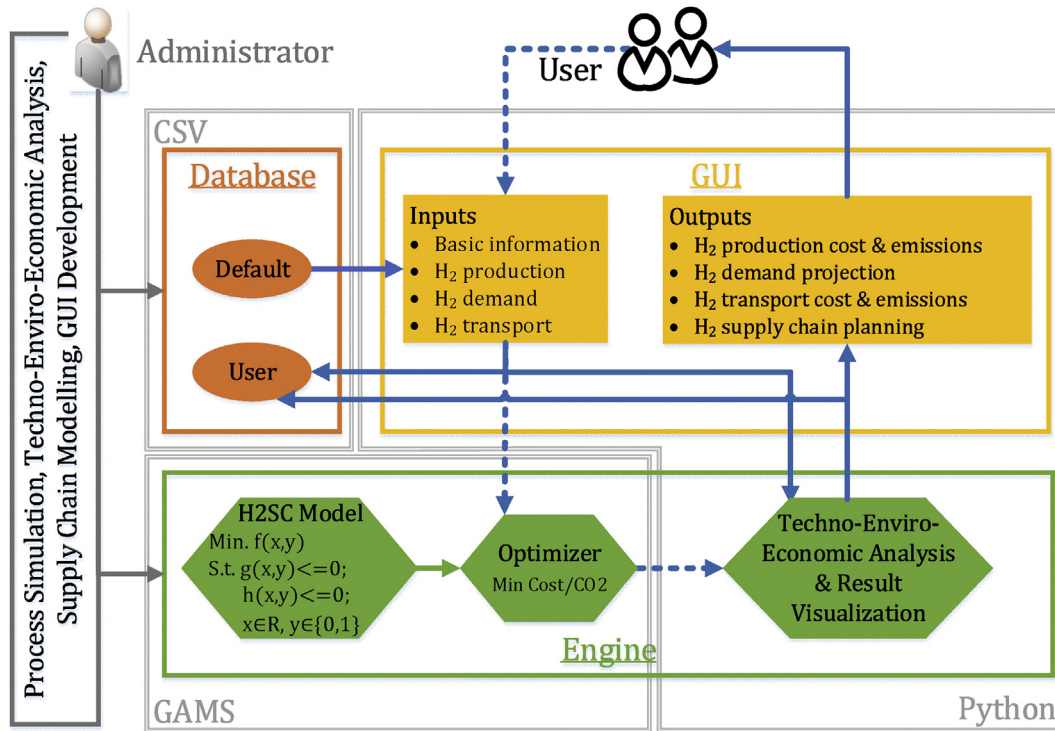


Fig. 1 – Framework of hydrogen economy assessment & resource tool (HEART).

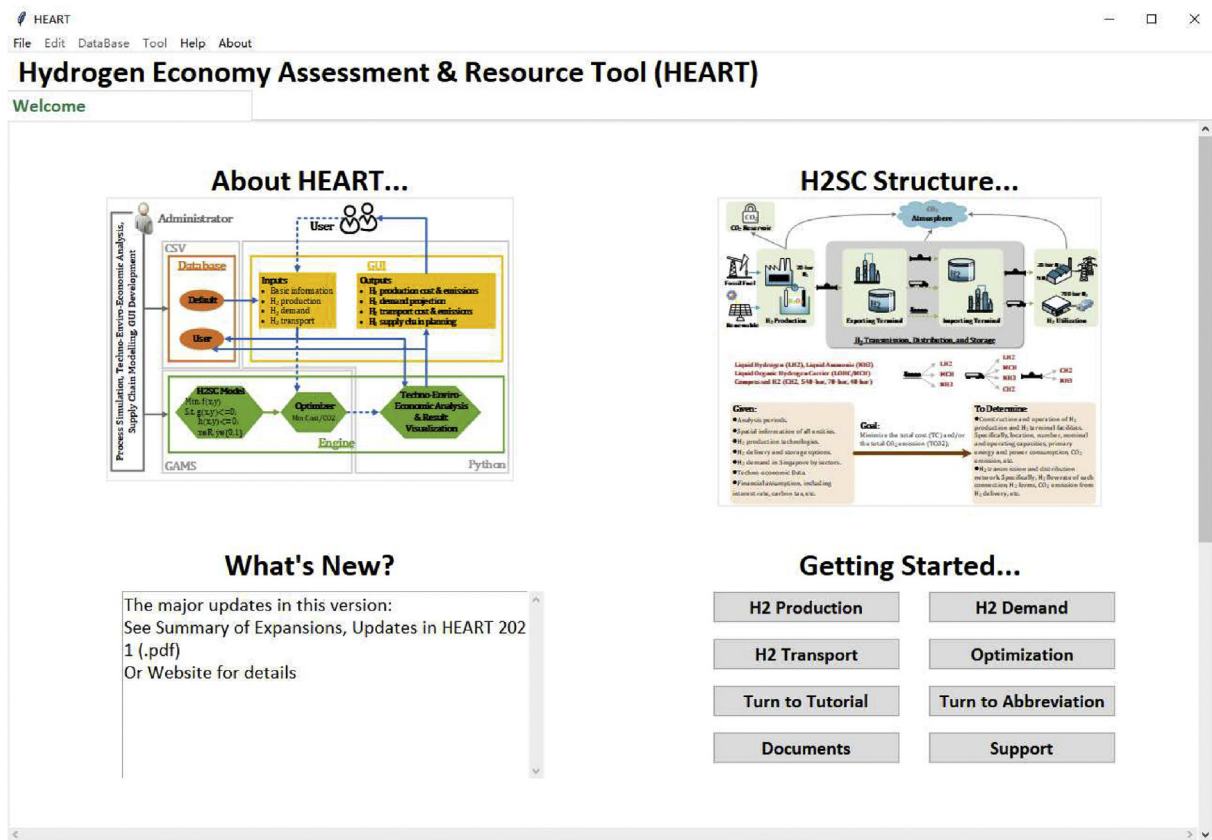


Fig. 2 – Homepage of the HEART graphical user interface.

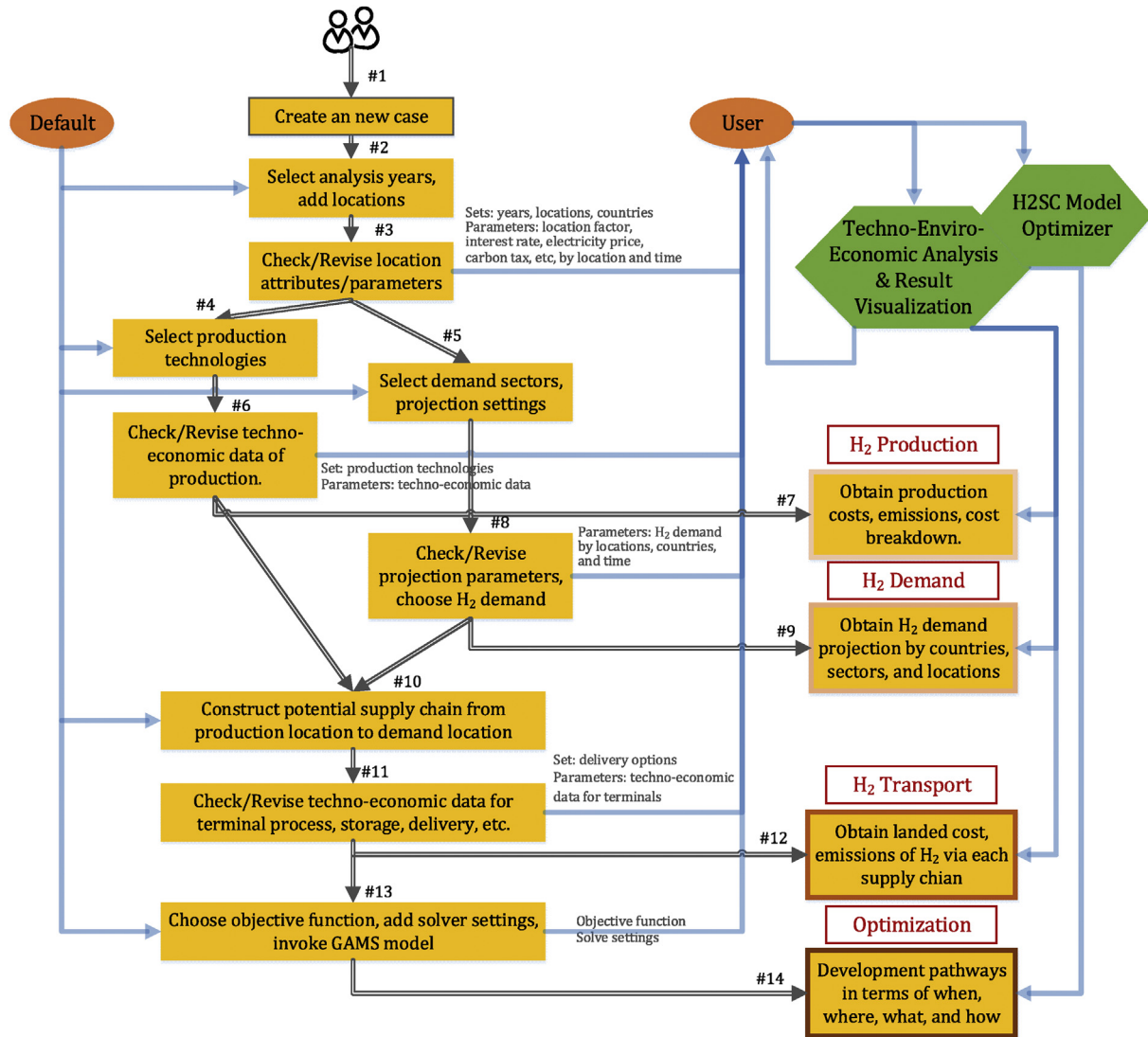


Fig. 3 – Workflow of a hydrogen supply chain project.

$$proj_{year} = a \times (year - year0) + base \times (1 + b)^{(year - year0)} \quad (1)$$

After choosing a model for change, the other parameter is set to zero. Both the fixed annual change (a) and the fixed rate (b) can be positive or negative. The value of a and b should depend on the scenarios users want to create. For example, 1%, 2% and 3% can be adopted to represent low, medium, and high increasing scenarios respectively. By adjusting the projection model parameters, one can create various scenarios, e.g., increasing carbon tax, increasing resource price, decreasing grid emissions factor, etc. Furthermore, users can also directly specify each attribute for the entire period if they want to set up scenarios which cannot be obtained by Eq. (1). After checking and/or revising the loaded data (#3), one can proceed to the next step.

Workflow – Step 2: H₂ production and H₂ demand

The next step is to select production technologies (#4) or demand sectors for the chosen locations (#5). Once a production

technology is selected for a location, the default techno-economic data is loaded, including capital cost, feedstock consumption, etc. Default data can be found in the [Supporting Information](#). Users can revise these inputs if needed (#6). A capital cost reduction factor and/or an efficiency improvement factor are adopted to consider the technological improvements in the future. Similar to the projection for location attributes, these two factors are projected independently via the two models discussed earlier and represented by the following general equation:

$$factor_{year} = a \times (year - year0) + (1 + b)^{(year - year0)} \quad (2)$$

The only difference between Eqs. (1) and (2) is that the base value in Eq. (2) is one. With the techno-economic data from the default database (see [Supporting Information](#)) or user inputs, H₂ cost, emissions, and cost breakdown are automatically obtained for all selected production technologies and displayed via both tables and figures (#7).

A critical factor in any H₂ economy model is the level of H₂ demand. In HEART, the H₂ demand is set exogenously. We

assume H_2 demand will be proportional to the total energy demand in each country. Therefore, the methodology to estimate H_2 demand relies on country-by-country projected energy demands and a H_2 penetration model. Energy projection profile over the analysis period can be obtained by two different methods: (1) using a regression equation from historical data [36] (the default database only includes ASEAN data, see [Supporting Information](#)); (2) manually provide a projection equation, similar with Eq. (1). H_2 penetration is assumed to follow an S-shaped logistic substitution model [37,38], as shown in Eq. (3),

$$N_t = \frac{K}{1 + e^{-\alpha t - \beta}} \quad (3)$$

where N_t and K represent the market penetration in year t and final market penetration (ceiling of growth) respectively; α specifies the steepness of the S-shape curve; β specifies the time to reach 50% K . In a practical application, the logistic equation, is also defined as:

$$N_t = \frac{K}{1 + e^{\left[\frac{\ln(81)}{\Delta t(t-t_m)} \right]}} \quad (4)$$

where, Δt and t_m represent the characteristic duration from 10% K to 90% K , and time to reach 50% K respectively. The profile shape can be changed by adjusting Δt and t_m . Smaller Δt and t_m will generate a faster H_2 generation growth market. Default values for Δt and t_m are both set as the half of the period of analysis, while K is assumed as 50%. Users can assign different demand projection parameters for each sector of each country, like power, industry, marine, domestic, and transport (#8). Furthermore, users can also directly assign H_2 demand of each year to create H_2 demand scenarios beyond the limitation of Eq. (4). Since total H_2 demand for each country is the sum of demand in various sectors, the total demand should be partitioned for different locations according to the proximity of different demand sectors. For example, after the H_2 demand for entire Malaysia is estimated, the next thing to decide is where the produced H_2 will be used. This will determine production planning at different locations and supply chain from production plants to end users (#9).

As shown in Fig. 3, one can obtain results about H_2 production (#5) and/or H_2 demand (#9) after the aforementioned steps. For further results regarding the landed cost of H_2 through a supply chain, one should proceed to construct the H_2 supply chain.

Workflow – Step 3: H_2 transport

On a H_2 transport page, a user constructs potential supply chains to deliver H_2 to all end users which have been previously selected when calculating H_2 demand (#10). As shown in Fig. 4, a supply chain starts with a production location and can include several terminals (e.g. export and import terminals) before reaching the final end-use location. For example, Fig. 4 shows a supply chain from Kerteh, Malaysia to Power Plants in Singapore. H_2 transport pathways and distances between terminals chosen for this example are labelled in Fig. 4. Pressurized hydrogen (20 bar H_2 or CH20P) is put in pipeline to take it from the production plant to a nearby terminal (distance

shown as 0 km). Liquefied ammonia sent by ship (LNH3S) from the Kerteh export terminal to an import terminal in Singapore. This means hydrogen is converted to liquid ammonia, stored for a specific period of time, and then loaded to a ship for onward transmission. The import terminal in Singapore (say located in Jurong Island) receives LNH3S, stores liquid ammonia (LNH3), decomposes LNH3, pressurizes the gaseous H_2 and distributes it locally via a pipeline at 40 bar pressure (CH40P). Finally, power plants receive H_2 from the pipeline network at 20 bar (CH20), which is lower due to an assumed fixed pressure drop along the transmission pipeline. The distance between locations should be provided by users, as shown in Fig. 4.

Once a supply chain is constructed and added to the user database, the techno-economic data for the related terminal processes, transmission methods, and distribution methods are automatically loaded from the default database and displayed in the GUI (#11). For example, the Jurong Island terminal receives LNH3S and delivers CH40P. This terminal is labelled as “Jurong Island – LNH3S – CH40P”. Then, techno-economic data for two processes (“LNH3S – LNH3” and “LNH3 – CH40P”) and one type of storage (“LNH3”) are loaded for Jurong Island customized using an appropriate location factor. Please refer to our published paper [22] for techno-economic data of H_2 transport.

In addition, the analysis tool will first automatically calculate the H_2 production cost and carbon footprint for the production location and find the H_2 demand for the particular end user, represented by red characters in Fig. 4. Then, it will calculate the landed cost and emissions of H_2 along the supply chain (#12). Here, results for a single year are calculated for the supply chain and estimated capacity without any optimization of the supply chain. Users can compare the landed cost and emissions by constructing different supply chains for different carrier options.

Workflow – Step 4: Optimization

The main goal of this study is a tool to construct a plausible hydrogen roadmap for decarbonization of the energy sector over a long-term period. Such a roadmap will answer the “4w questions” related to hydrogen infrastructure, namely when, where, what sizes and what technologies to build [39]. It is almost impossible for users to create all possible scenarios and compare them one by one using the previously described three steps. The tool consists of a superstructure-based supply chain optimization model which can find options with the minimum total cost or emissions based on user inputs. Users should select an objective function, such as total cost, and determine solver settings, such as the maximum allowed CPU time (#13). The total cost includes the cost of H_2 production, H_2 storage, H_2 delivery, H_2 terminal process, and H_2 end node, while the total CO_2 emission is attained by summing up all emissions from the consumed fossil fuel and electricity by production facilities, terminal processing facilities, delivery vehicles, and end nodes in the supply chain. It should be noted that carbon tax is also included in the total cost as operating cost for all processes. A brief introduction of the developed supply chain model can be found in the [Supporting Information](#).

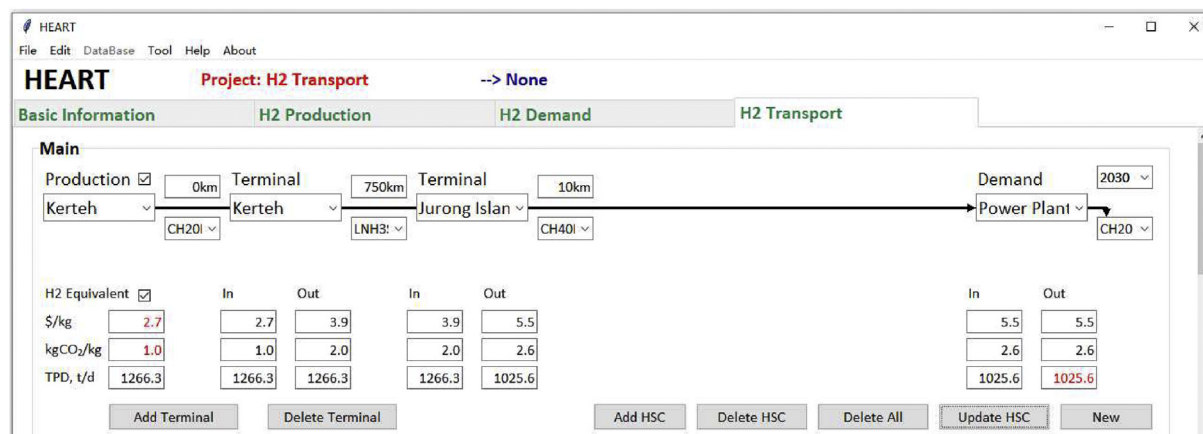


Fig. 4 – Snapshot of the HEART interface to construct a H₂ example Supply Chain.

With the selected production technologies, selected demand sectors (end-users), and the constructed supply chains, the tool will, first, automatically check at least one potential supply chain is constructed for each chosen demand. Then, the optimal alternative supply chains will be obtained by solving a MILP model. For example, if a user chooses one production technology in one location, one end user, and constructs supply chains via all possible carriers, the optimization model can find the best pathway (linked production and transport options) with the minimum total cost and/or emissions. Note that the assumptions of the case study depend on users' input and selections in the previous three steps.

The key elements of a H₂ roadmap, including but not limited to the landed cost and emission factors at different stages of the H₂ supply chain, the capital investment profiles, and H₂ delivery flow over the analysis period, can be obtained and visualized in suitable plots (#14). Furthermore, the carbon avoidance, the carbon avoidance cost [22], and the levelized cost of electricity (LCOE) can also be obtained based on the landed cost and emission factor of H₂. Visualization of results is handled by the Matplotlib library in Python. All key result data are stored in an Excel file, where each case is stored separately. A user can check results by choosing display options through the GUI. In addition, a user might get even more insightful conclusions from his/her own analysis.

Demonstration case study

Problem statement and assumptions

In this section, we create an ASEAN case study, where a supply chain among Singapore, Malaysia, Indonesia, and Thailand from 2021 to 2050 is constructed. We chose to minimize the total cost. The following simplifying assumptions are made to enable its use as an illustrative example:

- Singapore and Thailand are H₂ importers, while Malaysia and Indonesia are H₂ producers for internal consumption and export.

- In a producing country, there is only one production site and one export terminal. For example, Malaysia only produces H₂ in Kerteh and the terminal is in the same location.
- In an importing country, there is only one import terminal.
- The H₂ demand of each country is categorized by sectors, and the distribution distance from the import terminal to an end user is assumed for each category. For example, Indonesia-Power represents the H₂ demand for the power sector of entire Indonesia, and the distribution distance is assumed as 40 km. A simplified supply chain representation for the entire region including supply to the local demand centers in the producing countries is shown in Fig. 5.
- Coal gasification (CG) with CO₂ capture and storage (CCS) is considered as the potential H₂ production technology in Indonesia, while steam methane reforming (SMR) with CCS is the production technology adopted in Malaysia.
- Liquid ammonia is considered as the only hydrogen carrier for export.
- H₂ is distributed within countries via pipeline as 40-bar compressed gas (CH40P) to be received by the power, domestic (i.e. residential and commercial buildings), and industry sectors at 20-bar (CH20). For simplicity, these three sectors are considered as one, since they all use CH20 from a pipeline network. We use "PDI" prefixed with the country name to represent the combined demand of the three sectors in a country, e.g., Malaysia-PDI and Singapore-PDI. For road transport, H₂ is distributed within countries via trucks as 540-bar compressed gas (CH540T), and then compressed to 700-bar (CH700) when refueling vehicles. Liquid ammonia is distributed within countries via pipeline (LNH3P) for the marine sector, which will directly use liquid ammonia as marine fuel.
- The assumed target H₂ penetration levels in 2050 are 40% for Singapore, 30% for Malaysia and Thailand, and 20% for Indonesia.
- The location factor of Singapore, Malaysia, Thailand, and Indonesia are 1.20, 0.96, 1.00, and 0.71 respectively, which are obtained by the method from Sinnott R. and Towler G [35].

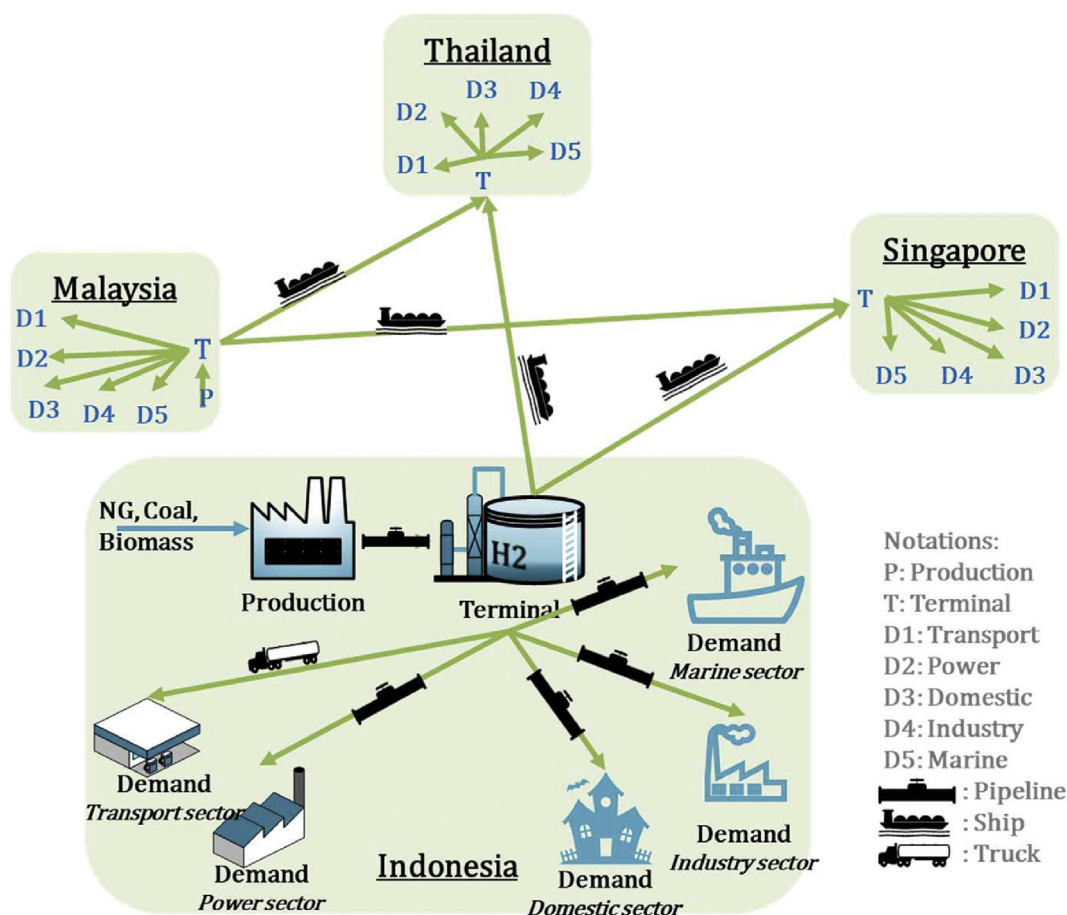


Fig. 5 – Simplified regional H₂ supply chain from Malaysia and Indonesia to Singapore and Thailand and their local supply chains within each country.

Questions to answer:

- How much H₂ should Singapore and Thailand import, and from where, over the period of 2021–2050?
- What is the landed cost of H₂ for each end-use in each country?
- What facilities are required for the exporters and importers? What is the capital investment for each facility?

Results and discussion

This optimization model requires all the steps discussed in [Section GUI and workflow of HEART](#). The steps are also shown in the video provided in the [Supporting Information](#). In the last step, we choose to optimize the total cost of the entire supply chain.

[Figs. 6–8](#) are directly obtained from HEART as outputs to answer the questions listed above. H₂ flows from the exporters to the importers are shown in [Fig. 6](#). Singapore and Thailand are predicted to import H₂ mostly from Bontang, Indonesia, as represented by orange and green lines. Thailand will increase the amount of liquid ammonia (LNH3) import from Malaysia after 2042. The main reason is the cost of LNH3 from Indonesia increases faster than the one from Malaysia

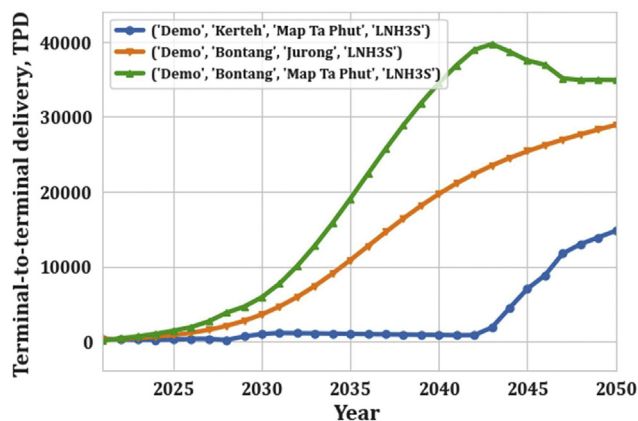
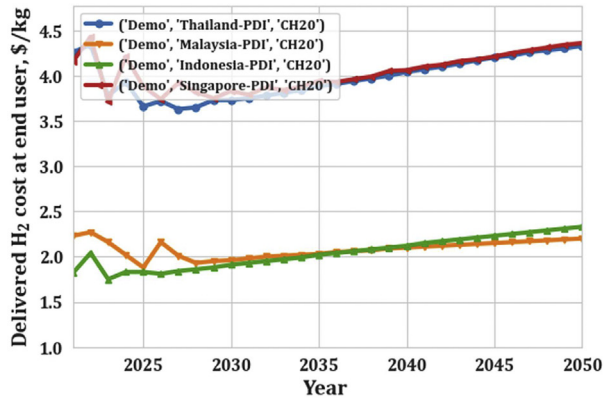
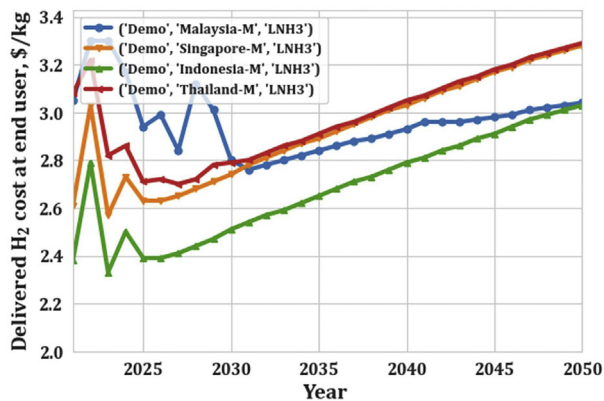


Fig. 6 – H₂ flow from exporters to importers. “Demo” is the project name. Kerteh, Jurong, Map Ta Phut, and Bontang are the terminals in Malaysia, Singapore, Thailand, and Indonesia respectively.

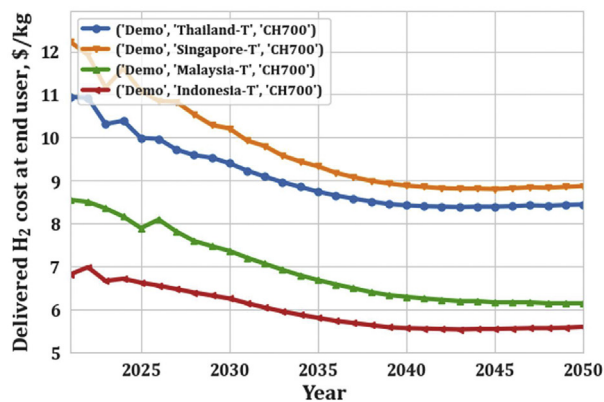
([Supporting Information Figure C1](#)), due to its higher carbon footprint ([Supporting Information Figure C2](#)) arising from higher grid emissions and higher emissions from H₂ production from coal in Indonesia. The shipping distance between



(a). CH20 cost represented as \$/kg H₂. "PDI" represents the power, domestic, and industry sectors in each country.



(b). LNH3 cost represented as \$/kg H₂ equivalent. "M" represents the marine sector in each country.



(c). CH700 cost represented as \$/kg H₂. "T" represents the road transport sector in each country.

Fig. 7 – H₂ cost for each sector in each country.

Indonesia and Thailand (3300 km) is also longer than the distance between Malaysia and Thailand (900 km). Despite the longer shipping distance, Indonesia initially enjoys lower hydrogen production costs and lower ammonia synthesis cost. The ammonia synthesis cost is lower in Indonesia due to lower location factor of capital cost and lower grid electricity

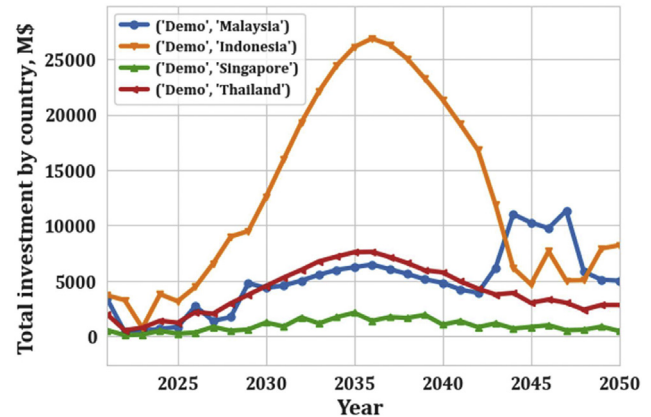


Fig. 8 – Capital Investment of each country from 2021 to 2050.

price. The higher cost of LNH3S via shipping from Indonesia including increasing carbon tax will eventually surpass 2.9 \$/kg H₂ equivalent to make import from Malaysia more cost effective.

In addition to the landed cost of CH20 for the PDI sectors, the costs of LNH3 for marine and CH700 for road transport are shown in Fig. 7. Broadly speaking, the delivered CH20 cost to the PDI users in Malaysia and Indonesia are comparable with the latter enjoying some initial cost advantage, as shown in Fig. 7(a). However, H₂ produced from CGCCS in Indonesia has a higher carbon footprint than H₂ produced from SMRCCS in Malaysia (Supporting Information Figure C3). Hence, H₂ production cost in Indonesia rises faster with increasing carbon tax and crosses over the cost to the PDI users in Malaysia after 2036. The delivered CH20 cost to the PDI users in Singapore and Thailand is almost two times of that in Malaysia and Indonesia. These extra costs are from shipping and cracking of LNH3. The latter one is the main contributor. The difference of shipping cost for bringing LNH3 to Thailand and Singapore is negligible. Thus, the delivered H₂ cost to the PDI users are practically the same.

Since LNH3 is directly used in the marine sector, its delivered cost to the marine uses in the four ASEAN countries under consideration are compared in Fig. 7(b) as \$/kg of H₂ equivalent. The delivered LNH3 cost in Malaysia is higher than in Indonesia because the latter enjoys the benefit from economy of scale due to a higher demand for marine fuel and export. The optimizer chooses to import LNH3 to Singapore and Thailand mostly from Indonesia. Hence, the cost pattern of LNH3 for direct use in the marine sector follows the same pattern.

The delivered cost of CH700 to the road transport sector shows an overall decreasing trend contributed by the assumed increasing utilization factor of refueling stations, as shown in Fig. 7(c). We assume that the utilization factor of refueling stations increases from 60% to 90% over the 30 year time period.

The total CO₂ delivery amount from two production locations is shown in the Supporting Information Figure C4, which also reflect the international and domestic H₂ demand trends in each country. It can be seen that the CO₂ delivery amount can

reach up to almost 2.5 million tonne per day for Bontang, Indonesia. Note that users can add constraints for CO₂ storage, e.g., a maximum storage capacity for a specific location.

Finally, the capital investment of each country from 2021 to 2050 is shown in Fig. 8. For Thailand and Singapore, the capital investment curves reflect their H₂ demand evolution. The capital investment curve of Singapore goes up and down because oversized infrastructures are constructed in a year that can accommodate demand growth for one additional year after which new investment is needed. The reason why other curves do not have similar phenomena is that their incremental H₂ demand per year is higher than Singapore. The incremental demand per year is large enough to benefit from the economy of scale. H₂ demand keeps rising before 2036 and then gradually reaches the penetration ceiling. Thus, the capital investment reaches a peak in 2036. As for Indonesia and Malaysia, their capital investment depends also on the import rate of other countries. The capital investment of Malaysia suddenly increases after 2042 consistent with an increase in export of LNH₃ from Malaysia shown in Fig. 6. Correspondingly, the capital investment of Indonesia decreases, as it becomes less attractive for Thailand to import LNH₃ from Indonesia.

Conclusions and outlook

In this paper, we introduce a multifunctional tool for H₂ roadmap studies, namely the Hydrogen Economy Assessment & Resource Tool or HEART. The tool is developed to be a versatile tool for both researchers and decision makers. This tool consists of three key components: (1) a comprehensive database for H₂ production, H₂ demand, and H₂ transport; (2) a versatile engine to carry out techno-enviro-economic analysis and supply chain optimization; (3) a graphical user interface (GUI) to easily set up case study scenarios to facilitate H₂ economy planning. Key elements related to H₂ economy are all included in the tool. And more importantly, with the developed case study workflow, analysis tool, and supply chain model, we can create various case studies to explore H₂ roadmap alternatives for full decarbonization.

A demonstration case study with some simplifying assumptions is presented to illustrate the capabilities of HEART including but not limited to estimating H₂ landed cost for various demand sectors, planning supply chains, etc. The tool can be used to analyze more complex and customized scenario settings and draw insightful conclusions regarding H₂ roadmaps for ASEAN or any country/region within ASEAN, e.g. develop blue H₂/green H₂ supply chain for ASEAN region, identify potential role of blue H₂/green H₂ in achieving National decarbonization target. The tool can also be applied for H₂ roadmap study for any regions across the world, with additional input. A broader study with the focus on a comprehensive ASEAN H₂ roadmap is in progress using HEART.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of abbreviations

AEL	alkaline electrolysis
ASEAN	Association of Southeast Asian Nations
ATR	autothermal reforming
BG	biomass gasification
CCS	CO ₂ capture and storage
CG	Coal gasification
CH ₂	compressed hydrogen
CH _{20P}	20 bar H ₂
CH _{40P}	40 bar H ₂ via pipeline
CH _{540T}	540 bar H ₂ via truck
CH ₇₀₀	700 bar H ₂
GAMS	General Algebraic Modeling System
GHG	greenhouse gas
GUI	graphical user interface
HEART	Hydrogen Economy Assessment & Resource Tool
LCOE	levelized cost of electricity
LH ₂	liquid hydrogen
LNH ₃	liquid ammonia
LNH _{3P}	liquid ammonia via pipeline
LNH _{3S}	liquid ammonia via ship
LOHC	liquid organic hydrogen carrier
MCH	methylcyclohexane
MILP	mixed-integer linear programming
MOF	metal–organic framework
PEM	proton exchange membrane electrolysis
PDI	power, domestic, and industry sectors
POX	partial oxidation
SMR	steam methane reforming
SOEL	solid oxide electrolysis
WE	water electrolysis

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2022.05.036>.

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